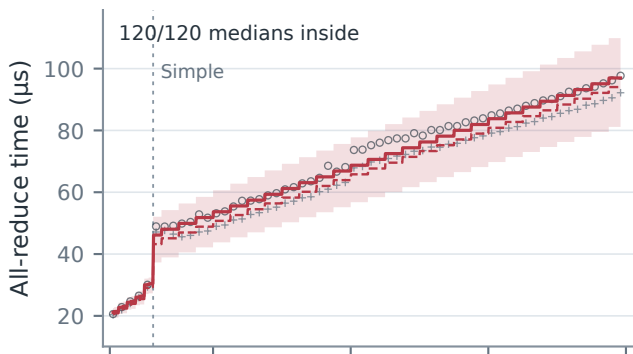


NCCL protocol work explains the timing step

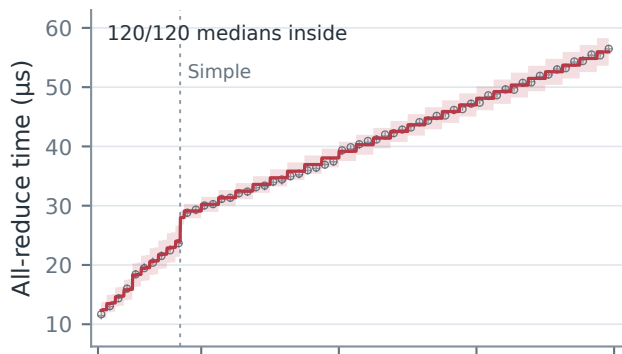
Independent validation | 60 new payloads per curve | Parameters locked before capture

— Model: GPU events Software + method band + Benchmark measurement
- - - Model: benchmark ○ GPU-event measurement × Outside band

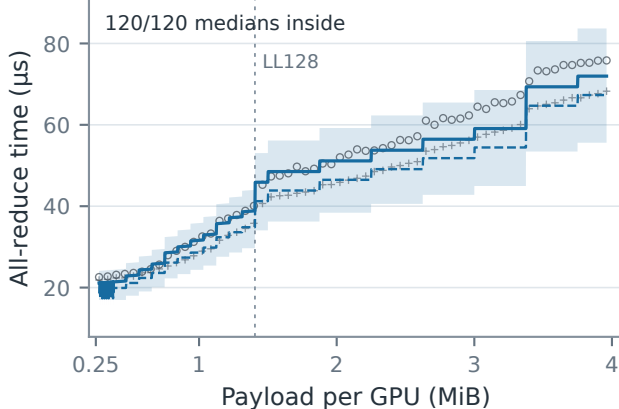
A100 | 2 GPUs



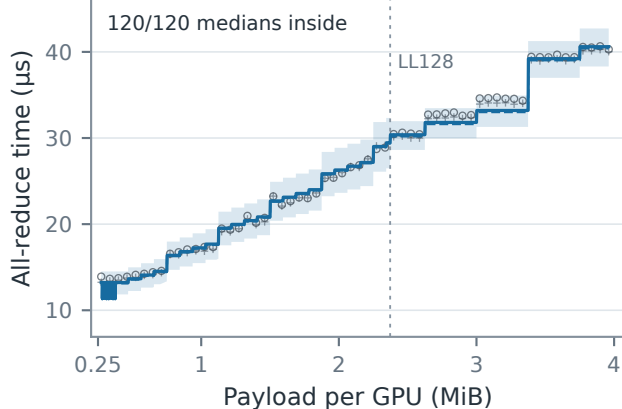
GH200 | 2 GPUs



A100 | 4 GPUs



GH200 | 4 GPUs



Band: calibration residuals, repeat spread and method allowance. Gold crosses mark uncovered medians.

LL carries data and flags together; Simple fences data, publishes counters, and polls in the same kernel.